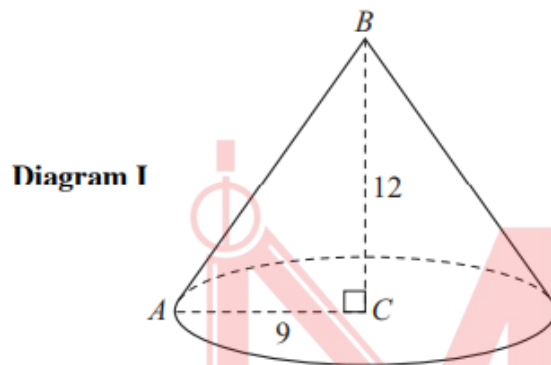


Mensuration

Worksheet 2c

Cone

1. S10/qp22/q12



[Volume of a cone = $\frac{1}{3}\pi r^2 h$]

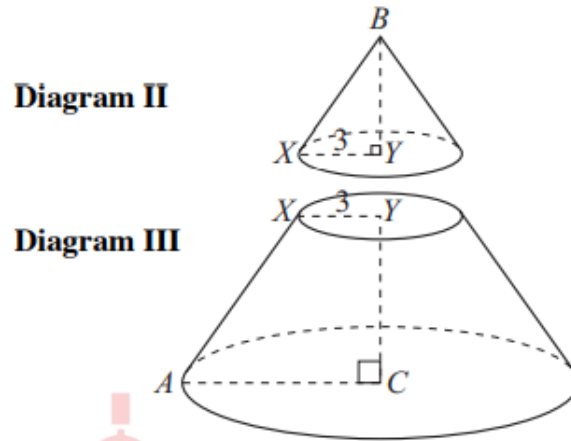
[Curved surface area of a cone = $\pi r l$]

Diagram I shows a solid cone with C as the centre of its base. B is the vertex of the cone and A is a point on the circumference of its base. $AC = 9$ cm and $BC = 12$ cm.

a. Calculate

- AB , [2]
- the total surface area of the cone, [2]
- the volume of the cone. [2]

- b. The cone in Diagram I is cut, parallel to the base, to obtain a small cone shown in Diagram II and a frustum shown in Diagram III. Y is the centre of the base of the small cone. X is the point on the circumference of this base and on the line AB such that Diagram III $XY = 3$ cm.



Calculate

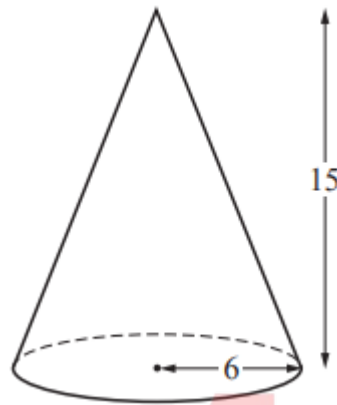
- | | | |
|------|--|-----|
| i. | BY, | [1] |
| ii. | AX, | [1] |
| iii. | the circumference of the base of the small cone, | [2] |
| iv. | the volume of the frustum. | [2] |

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2. S14/qp22/q9

[Volume of a cone = $\frac{1}{3}\pi r^2 h$]

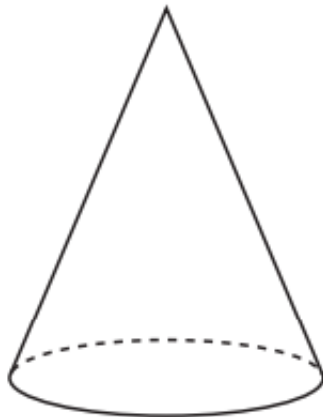
[Curved surface area of a cone = $\pi r l$]



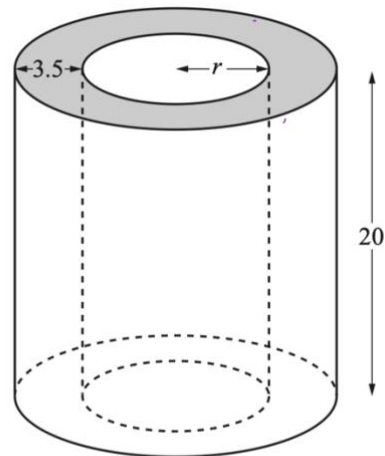
The diagram shows a solid cone of height 15cm and base radius 6cm.

- a. Show that the slant height of the cone is 16.2cm, correct to one decimal place. [1]
- b. Calculate the total surface area of the cone. [3]
- c. Calculate the volume of the cone. [2]
- d. The cone is made from wood. The mass of $1m^3$ of the wood is 560 kg. Calculate the mass of the cone in grams. [2]
- e. Another cone is made of the same material and is geometrically similar to the first. The mass of the second cone is double the mass of the first.
 - i. Calculate the height of the second cone. [2]
 - ii. Calculate the total surface area of the second cone. [2]

3. S16/qp22/q11a(ii)



Solid II



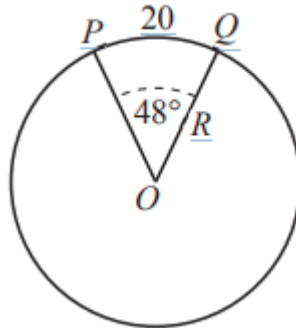
Solid I

Solid II is a cone with volume of 3000cm^3 . The perpendicular height of the cone is twice its radius. Which solid is the taller and by how much? [4]

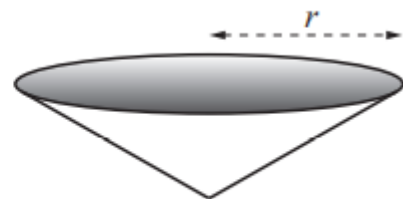
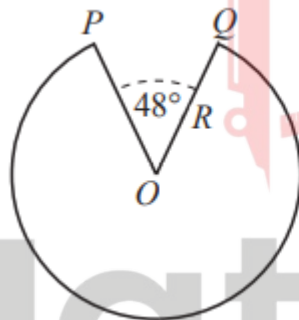
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4. W16/qp22/q8

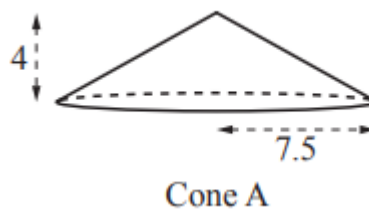
- a. P and Q are points on the circumference of a circle, centre O, radius R cm. The minor arc PQ = 20cm and $\angle POQ = 48^\circ$.
- i. Show that $R = 23.9$, correct to one decimal place. [3]



- ii. Calculate the area of the minor sector POQ. [2]
- iii. The minor sector POQ is removed from the circle and the remaining major sector is shaped to form an open cone of radius r cm. Calculate r. [2]



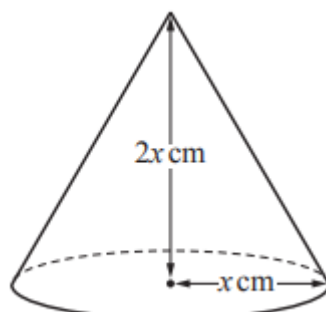
- b. [The curved surface area of a cone is $\pi r l$, where l is the slant height]
Cone A has radius 7.5 cm and height 4cm.



- i. Calculate the curved surface area of cone A. [3]
- ii. Cone B is geometrically similar to cone A. The ratio curved surface area of cone A : curved surface area of cone B is 64 : 25 . Find the height of cone B. [2]

5. W16/qp41/q3(b) 0580

[Volume of a cone = $\frac{1}{3}\pi r^2 h$]

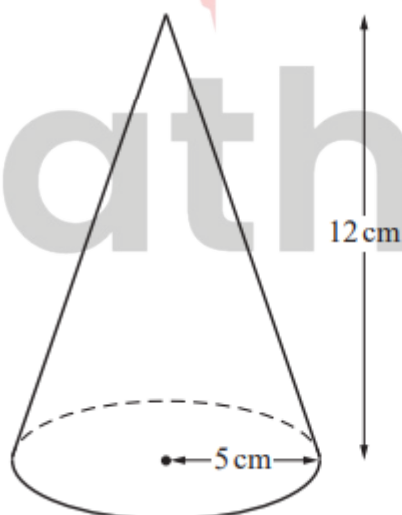


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The cone in the diagram has radius $x \text{ cm}$ and height $2x \text{ cm}$. The volume of the cone is 500 cm^3 . Find the value of x . [3]

6. W17/qp42/q2(b) 0580

The curved surface area, A , of a cone with radius r and slant height l is $A = \pi r l$



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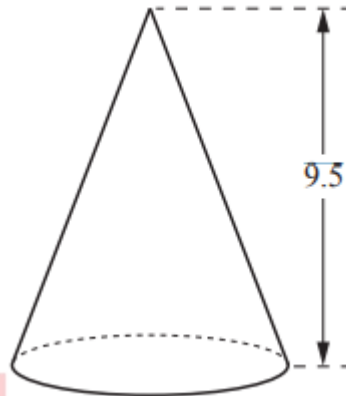
The diagram shows a solid cone with radius 5 cm and perpendicular height 12 cm .

- i. The total surface area is painted at a cost of $\$0.015 \text{ per cm}^2$.
Calculate the cost of painting the cone. [4]
- ii. The cone is made of metal and is melted down and made into smaller solid cones with radius 1.25 cm and perpendicular height 3 cm .
Calculate the number of smaller cones that can be made. [3]

7. S18/qp22/q5(b)

[Volume of a cone = $\frac{1}{3}\pi r^2 h$]

[Curved surface area of a cone = $\pi r l$]

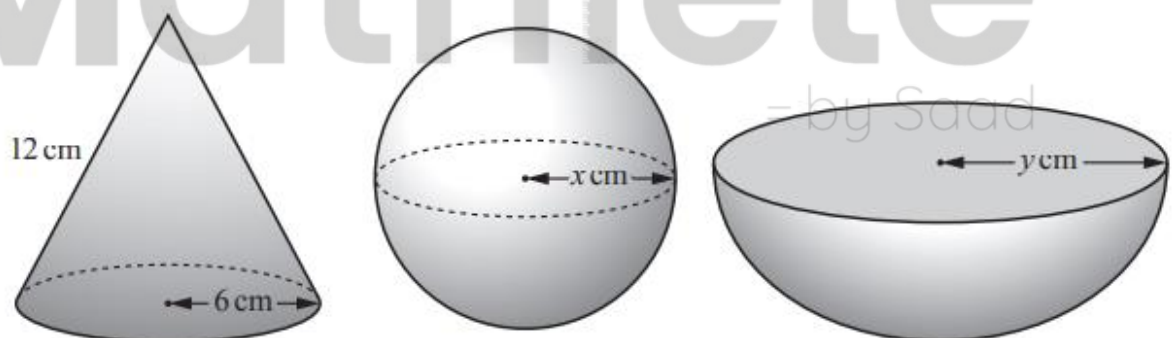


A cone has height 9.5 cm and volume 115 cm³.

- i. Show that the radius of the base of the cone is 3.4 cm, correct to 1 decimal place. [2]
- ii. Calculate the curved surface area of the cone. [3]

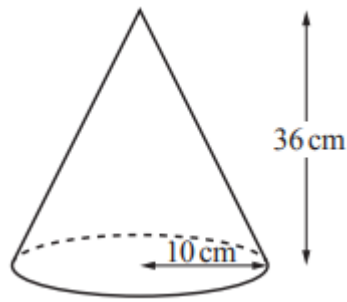
8. S18/qp43/q7(b)(i) 0580

[The curved surface area, A , of a cone with radius r and slant height l is $A = \pi r l$]



The diagram shows three solids. The base radius of the cone is 6 cm and the slant height is 12 cm. The radius of the sphere is x cm and the radius of the hemisphere is y cm. The total surface area of each solid is the same. Show that the total surface area of the cone is 108 cm².

9. W18/qp42/q10(b)(i) 0580

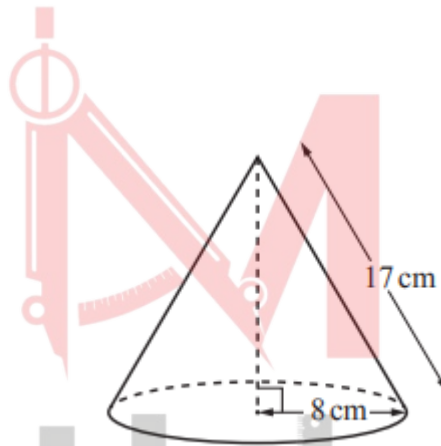


A solid metal cone has radius 10cm and height 36cm.

Calculate the volume of this cone.

[2]

10. W18/qp43/q3(a) 0580



The diagram shows a solid cone.

The radius is 8 cm and the slant height is 17cm.

- Calculate the curved surface area of the cone.
- Calculate the volume of the cone.
- The cone is made of wood and 1cm^3 of the wood has a mass of 0.8 g.

Calculate the mass of the cone.

- The cone is placed in a box.
The total mass of the cone and the box is 1.2kg.
Calculate the mass of the box.
Give your answer in grams.

[2]

[4]

[1]

[1]

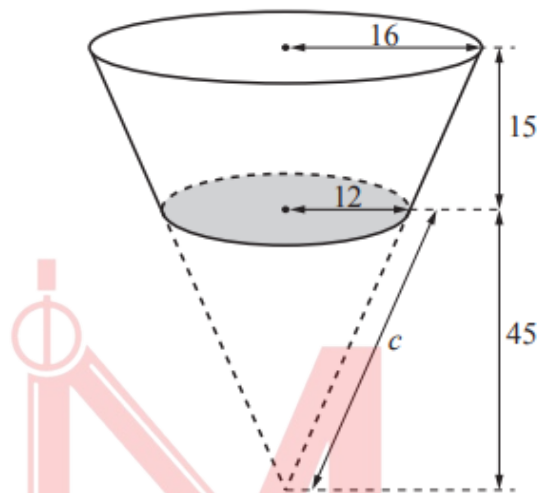
11. M19/qp22/q12 0580

A cone with height 14.8cm has volume 275cm^3 .

Calculate the radius of the cone.

[3]

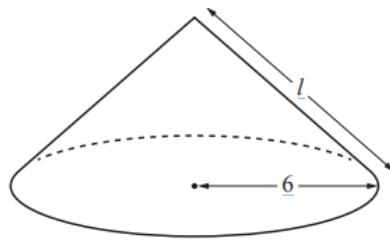
12. W19/qp22/q8



The diagram shows a bowl with a circular base. The curved surface of the bowl is formed by removing a cone with radius 12cm and height 45cm from a larger cone as shown in the diagram. The radius of the top of the bowl is 16cm and its height is 15cm .

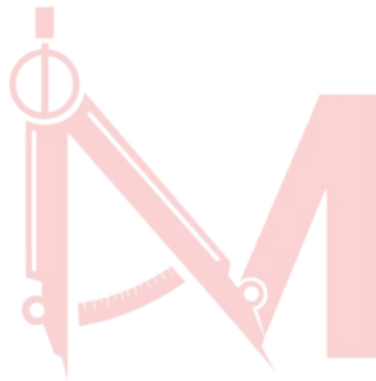
- Calculate the volume of the bowl. [3]
- The slant height of the cone that has been removed is c cm. Show that $c = 46.6$, correct to 3 significant figures. [2]
- The bowl is completely filled with water. Calculate the total surface area of the bowl that is in contact with the water. [4]

13. S21/qp22/q9



A cone has radius 6 cm and slant height l cm. The total surface area of the cone is 84 cm^2 .

- a. Show that $l = 8$. [2]
- b. Calculate the volume of the cone. [3]
- c. A similar cone has a total surface area of 47.25 cm^2 . Find the radius of this cone. [2]



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Answer Key

1. S10/qp22/q12

12	(a) (i)	15	P2	After P0, $\sqrt{9^2 + 12^2}$ P1
	(ii)	$678 - 679 \text{ (cm}^2\text{)}$	S2	After S0, $\pi \times 9 \times \text{their } 15 + \pi \times 9^2$ S1
	(iii)	$1017 - 1020 \text{ (cm}^3\text{)}$	V2	After V0, $\frac{1}{3} \times \pi \times 9^2 \times 12$ V1
	(b) (i)	4 cm	B1	
	(ii)	10 cm	B1	
	(iii)	$18.8 - 18.9 \text{ (cm)}$	C2	After C0, $\pi \times 3 \times 2$ C1
	(iv)	$979 - 983 \text{ (cm}^3\text{)}$	W2	After W0, $\frac{26}{27} \times \text{their } 1018$ or
			[12]	their $1018 - \frac{1}{3} \pi 3^2 \times \text{their } 4$ W1

2. S14/qp22/q9

9	(a)	$\sqrt{15^2 + 6^2} = 16.15(5\dots)$	1	Must be shown to at least 2 d.p.
	(b)	417 to 419	3	M1 for $\pi \times 6 \times 16.2$ soi by 305.4 M1 for $\pi \times 6^2$ soi by 113.1
	(c)	565 to 566	2	M1 for $\frac{1}{3} \times \pi \times 6^2 \times 15$ or better
	(d)	316 to 317	2FT	FT <i>their</i> (c) $\times 0.56$ evaluated B1 for figs 316(...) or 317(...) or <i>their</i> (c) $\times$ figs 56 evaluated
	(e) (i)	18.89 to 18.9	2	M1 for $\sqrt[3]{2}$ or 1.25... seen oe
	(ii)	662 to 665	2	M1 for $(\sqrt[3]{2})^2$ or 1.58... seen oe

3. S16/qp22/q11a(ii)

(ii)	Solid II by 2.5 – 2.6	4	B3 11.25 to 11.3 cm or M1 for $\frac{1}{3} \times \pi r^2 \times 2r = 3000$ or better and M1 for $r^3 = \frac{3000 \times 3}{2 \times \pi}$ (= 1432)
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4. W16/qp22/q8

8	(a) (i)	Dependent on 4 fig. term calculated using any version of π .	3	M1 for arc length $\frac{48}{360} \times 2\pi R$ soi and M1 for $R = 20 \times \frac{360}{48} \times \frac{1}{2\pi}$ oe
	(ii)	239	2	M1 for $\frac{48}{360} \times \pi R^2$
	(iii)	20.7	2	M1 for $2\pi r = \frac{312}{360} \times 2\pi R$ oe
	(b) (i)	200	3	M1 for $l^2 = 4^2 + 7.5^2$ oe soi and A1 for $(l =) 8.5$
	(ii)	2.5	2	B1 for 8 : 5 soi

5. W16/qp41/q3(b) 0580

(b)	6.2[0] or 6.203 to 6.204	3	M2 for $x^3 = \frac{500}{\frac{2}{3}\pi}$ oe or better or M1 for $\frac{1}{3} \times \pi \times x^2 \times 2x = 500$ oe
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6. W17/qp42/q2(b) 0580

2(b)(i)	4.24 or 4.241 to 4.242	4	M3 for $(\pi \times 5^2 + \pi \times 5 \times \sqrt{5^2 + 12^2})$ or M2 for $\pi \times 5 \times \sqrt{5^2 + 12^2}$ or M1 for $5^2 + 12^2$ or $\pi \times 5^2$
2(b)(ii)	64 cao final answer	3	M2 for $\frac{[k\pi] \times 5^2 \times 12}{[k\pi] \times 1.25^2 \times 3}$ or M1 for $\frac{1}{3} \times \pi \times 5^2 \times 12$ or $\frac{1}{3} \times \pi \times 1.25^2 \times 3$ OR M2 for 4^3 or $\left(\frac{1}{4}\right)^3$ seen or M1 for factor 4 or $\frac{1}{4}$ soi

7. S18/qp22/q5(b)

5(b)(i)	$\frac{1}{3}\pi r^2 \times 9.5 = 115$ or $r^2 = \frac{3V}{\pi h}$ or better	M1	Correct substitution into volume equation or correct rearrangement
	$r = 3.39[9\dots]$ or $3.40[00]$	A1	
5(b)(ii)	108 or 107.7 to 107.8	3	M2 for $\pi \times 3.4 \times \sqrt{9.5^2 + 3.4^2}$ or M1 for $l^2 = 9.5^2 + 3.4^2$ soi

8. S18/qp43/q7(b)(i) 0580

7(b)(i)	$\pi \times 6 \times 12 + \pi \times 6^2 = 108\pi$	M2	M1 for $\pi \times 6 \times 12$
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9. W18/qp42/q10(b)(i) 0580

10(b)(i)	3770 or 3769 to 3770....	2	M1 for $\frac{1}{3} \times \pi \times 10^2 \times 36$
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10. W18/qp43/q3(a) 0580

3(a)(i)	427 or 427.2 to 427.3...	2	M1 for $\pi \times 8 \times 17$
3(a)(ii)	1010 or 1005....	4	M2 for $\sqrt{17^2 - 8^2}$ oe or M1 for $h^2 + 8^2 = 17^2$ oe M1 for $\frac{1}{3} \times \pi \times 8^2 \times \text{their } h$ oe
3(a)(iii)	804 or 804.2 to 804.4 or 808	1	FT their (ii) $\times 0.8$
3(a)(iv)	396 or 395.6 to 395.8 or 392	1	FT 1200 – their (iii)

11. M19/qp22/q12 0580

12	4.21 or 4.212....	3	M2 for $\sqrt{\frac{275 \times 3}{14.8 \times \pi}}$ oe or M1 for $275 = \frac{1}{3} \times \pi \times r^2 \times 14.8$ oe
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12. W19/qp22/q8

8(a)	9300 or 9299 to 9301	3	M2 for $\frac{1}{3}\pi \times 16^2 \times 60 - \frac{1}{3}\pi \times 12^2 \times 45$ oe or M1 for $\frac{1}{3}\pi \times 16^2 \times 60$ or $\frac{1}{3}\pi \times 12^2 \times 45$
8(b)	$[c^2 =] 12^2 + 45^2$	M1	
	$[c =] 46.57\dots$	A1	
8(c)	1820 or 1816 to 1819.[0...]	4	B2 for $l = 62.09$ to 62.13 or M1 for $\sqrt{60^2 + 16^2}$ oe and M1 for $\pi \times 16 \times \text{their } 62.1 - \pi \times 12 \times 46.6$ [$+\pi \times 12^2$] If 0 scored, SC1 for $\pi \times 12^2$

13. S21/qp22/q9

9(a)	$6\pi l + \pi \times 6^2$ OR $84\pi - \pi \times 6^2$	M1	
	$6\pi l + \pi \times 6^2 = 84\pi$ leading to $l = 8$ OR $6\pi l = 84\pi - \pi \times 6^2$ leading to $l = 8$	A1	A0 if any errors or omissions
9(b)	199 or 200 or 199.4 to 199.5...	3	M2 for $\frac{1}{3}\pi \times 6^2 \times \sqrt{8^2 - 6^2}$ or M1 for $8^2 - 6^2$ or $\frac{1}{3}\pi \times 6^2 \times \text{their } h$
9(c)	4.5 nfw	2	B1 for $\sqrt{\frac{47.25\pi}{84\pi}}$ soi or $\sqrt{\frac{84\pi}{47.25\pi}}$ soi or M1 for a correct equation in r